

From condn 2, Row 2 Column 1 is 0. By condition 3, row 2 column 2 is 0. The corresponding matrix is $\begin{bmatrix} 0 & \square \\ 0 & 0 \end{bmatrix}$

• Row 1 must have a leading entry, because otherwise there is a possibility of Row 1 being zero and Row 2 being non-zero, which violates Condition 1.

6) Repeat Exercise 5 for a nonzero 3×2 matrix.

Soln: We shall use the conditions listed in Problem 5 which a matrix in echelon form must satisfy.

• Row 1 must have a pivot ($\square$), since o.w. the matrix being nonzero, there is a possibility of Row 1 being a zero row. In that situation, at least one row of 2 or 3 must be non-zero. This violates Condition 1.

• By Condition 3, Row 2 Column 1 = Row 3 Column 1 = 0 if we assume Row 1 has the pivot in column 1.

• Let Row 2 have a pivot ($\square$). Given the constraint of 2 columns and applying Condition 2, Row 2 Column 2 = $\square$. $\therefore$ By Condition 3, Row 3 Column 2 = 0

• The corresponding matrices are: $\begin{bmatrix} \square & 0 \\ 0 & \square \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \square & * \\ 0 & \square \\ 0 & 0 \end{bmatrix}$

• Suppose, it's not necessary that Row 2 has a pivot. Then, the third row must be a zero row since there is a possibility that Row 2 can be zero row, which otherwise will violate condition 1

• The corresponding matrices are: $\begin{bmatrix} \square & 0 \\ 0 & * \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \square & * \\ 0 & * \\ 0 & 0 \end{bmatrix}$

• Let, Row 1 has the leading entry in Column 2. By defn. of leading entry, Row 1 Column 1 = 0. By Condition 3, Row 2 Column 2 = Row 3 Column 2 = 0